IEEE JOURNAL OF

BIOMEDICAL AND HEALTH INFORMATICS

J-BHI Special Issue on "AI-Driven Multimodal Intelligence for Sustainable Healthcare Systems"

Multimodal AI integrates diverse data modalities, including Electronic Health Records (EHRs), imaging, speech, genomic data, wearable sensor streams, and environmental inputs. By learning from cross-domain representations, multimodal models can achieve context-aware inference and holistic patient understanding, supporting healthcare systems that are both intelligent and adaptive. These capabilities make MMAI a foundational enabler for sustainable healthcare systems, which emphasize efficiency, inclusivity, and resilience over time. In contrast to short-term optimization, sustainable healthcare leverages AI to ensure long-term medical accessibility, equitable resource allocation, and responsible data usage. The integration of multimodal AI into healthcare promotes the emergence of AI-driven sustainable ecosystems capable of addressing global challenges such as aging populations, chronic diseases, and environmental health threats. Through technologies such as federated learning, edge intelligence, blockchain-based data security, and explainable AI, these systems can perform secure, privacy-preserving, and interpretable medical analytics. Moreover, AI-powered decision support can optimize hospital energy consumption, improve logistics in healthcare supply chains, and reduce the environmental footprint of clinical operations—linking digital transformation to the goals of sustainable development.

This special issue, "AI-Driven Multimodal Intelligence for Sustainable Healthcare Systems," seeks to consolidate cutting-edge research, case studies, and technological developments that explore how AI can transform healthcare into a more sustainable, human-centric, and ethically grounded domain. The issue provides a multidisciplinary platform bridging computer science, biomedical engineering, cognitive computing, environmental science, and health policy, encouraging collaboration among researchers, clinicians, and technologists. The general justification for this special issue lies in the urgent global demand for healthcare systems that are not only intelligent but also sustainable. As AI continues to evolve toward strong and general intelligence, its responsible integration into healthcare becomes a critical research priority. Multimodal AI, with its ability to combine heterogeneous data and enable contextual reasoning, represents the cornerstone of this transformation. The special issue thus aims to highlight innovative methodologies, scalable architectures, and ethical frameworks that can accelerate the realization of sustainable, inclusive, and trustworthy healthcare systems powered by AI.

Topics of interest include, but are not limited to, the following:

- Development and optimization of Multimodal Large Language Models (MLLMs)
- Multimodal data fusion frameworks
- Integration of MLLMs
- Predictive analytics and intelligent modeling
- AI-driven remote monitoring and telemedicine
- Human–AI interaction and patient–doctor communication modeling
- Medical image understanding and generation
- Benchmark datasets, evaluation protocols, and clinical validation
- Explainable AI (XAI)
- Data privacy, security, and federated learning approaches
- Ethical, legal, and regulatory frameworks
- Fairness, accountability, and bias mitigation strategies

Guest Editors

Mu-Yen Chen, National Cheng Kung University, Taiwan, mychen119@gs.ncku.edu.tw
Victor Leung, University of British Columbia, Canada, vleung@ece.ubc.ca
Vincenzo Piuri, University of Milan, Italy, vincenzo.piuri@unimi.it
Patrick C. K. Hung, Ontario Tech University, Canada, Patrick.Hung@ontariotechu.ca
Marcelo Fantinato, University of São Paulo, Brazil, m.fantinato@usp.br

Key Dates

Deadline for Submission: 31 October, 2026
First Reviews Due: 30 November, 2026
Revised Manuscript Due: 31 December, 2026
Revised Manuscript Due: 31 January, 2027

